

# Claude Vibe Coding & Engineering

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## A Complete Guide

5 chapters compiled into one PDF

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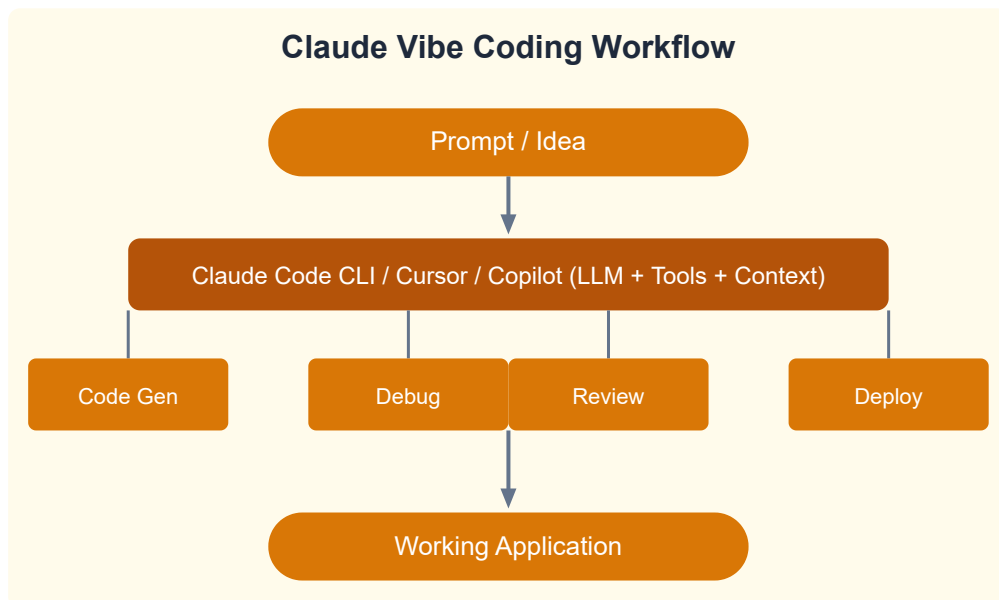
**Please do not print this document unnecessarily.** Think of the environment — save paper and trees. Share the digital link instead:  
<https://snehaliteng.github.io/tutorials/claude-vibe-course/>

# Vibe Coding for Fun & Profit

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## Core Concepts

"Vibe coding" is an AI-native coding paradigm where you describe what you want in natural language and let AI generate the code. The shift is from writing code line-by-line to curating AI-generated output.



# AI Foundations

| Concept             | Description                                                                         |
|---------------------|-------------------------------------------------------------------------------------|
| LLMs                | Large Language Models power code generation (Claude, GPT-4o)                        |
| Tools & Loops       | AI reads files, runs commands, iterates on feedback                                 |
| Context Engineering | Writing effective prompts and files <code>agents.md</code> / <code>claude.md</code> |
| Workflow Evolution  | From manual coding to AI-assisted to fully AI-driven                                |

# IDE Hands-On

Compare the major AI coding IDEs:

| Tool        | Strengths                              | Best For           |
|-------------|----------------------------------------|--------------------|
| Cursor      | Deep context awareness, inline editing | Daily development  |
| Copilot     | GitHub integration, VS Code native     | Enterprise teams   |
| Codex CLI   | Terminal-native, agentic actions       | Automation scripts |
| Claude Code | Full project context, checkpoints      | Complex codebases  |

## Building a Kanban App

```
# In Cursor or Claude Code CLI, you can prompt:
"Create a full-stack Kanban board with FastAPI backend and vanilla JS frontend.
Include: drag-and-drop, add/delete columns, persist to SQLite."

# The AI will generate:
# backend/main.py - FastAPI routes
# frontend/index.html - Drag-and-drop UI
# backend/models.py - SQLite models
```

## Multi-Model Environments

- **YOLO Mode** — Let AI run commands autonomously
- **Responsible Coding** — Always review AI-generated code for security
- **Cross-Model Collaboration** — Switch between Claude, GPT-4o, and local models

## Full-Stack Workflows

Build commercial MVPs quickly: FastAPI + Docker + SQLite + a modern frontend.

```
# FastAPI setup
pip install fastapi uvicorn sqlalchemy

# Dockerfile
FROM python:3.11
COPY . /app
RUN pip install -r requirements.txt
CMD ["uvicorn", "main:app", "--host", "0.0.0.0"]
```

 **Exercise:** Use Claude Code CLI or Cursor to build a Kanban board app. Start with "Create a Kanban board with FastAPI backend and HTML/CSS/JS frontend." Iterate by adding drag-and-drop, local persistence, and column management. Use Docker to containerize the final product.

# Vibe Engineering as a Professional

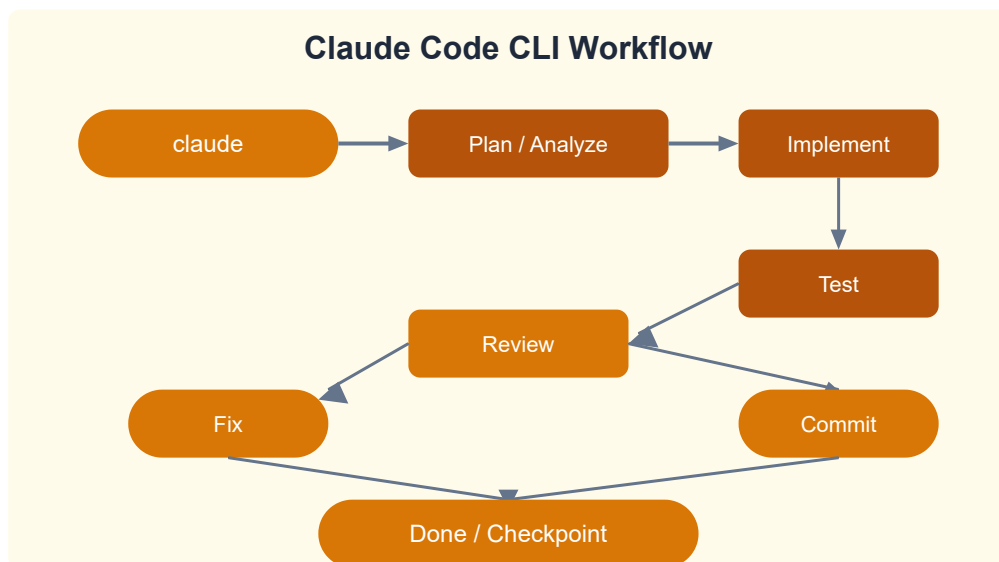
## Claude Code CLI

Claude Code is Anthropic's agentic CLI tool. It operates in your terminal with full file system access, git integration, and the ability to run commands.

```
# Installation
npm install -g @anthropic-ai/claude-code

# Start a session
claude

# Or pipe a prompt directly
echo "Explain this codebase" | claude -p
```



## Advanced Controls

| Command                  | Description                                     |
|--------------------------|-------------------------------------------------|
| <code>/plan</code>       | Generate a plan before writing code             |
| <code>/review</code>     | Review recent changes                           |
| <code>/checkpoint</code> | Save a session checkpoint                       |
| <code>--yolo</code>      | Autonomous mode — run commands without approval |

```
# Using checkpoints
claude --checkpoint save "pre-refactor"
# ... make changes ...
claude --checkpoint load "pre-refactor" # Rollback if needed
```

## Ecosystem Extensions: MCP Servers

Model Context Protocol (MCP) servers extend Claude's capabilities with custom tools.

```
// MCP server configuration in claude.json
{
  "mcpServers": {
    "jira": {
      "command": "node",
      "args": ["mcp-jira-server.js"],
      "env": { "JIRA_TOKEN": "${JIRA_TOKEN}" }
    },
    "github": {
      "command": "node",
      "args": ["mcp-github-server.js"]
    }
  }
}
```

## Professional Integrations

- **Jira** — Create and update tickets from Claude
- **GitHub** — PR review, issue management, code review
- **Custom Skills** — Reusable prompt templates for common tasks

## Production SaaS Project

Build a legal document generator SaaS with:

- `claude.md` project context file
- Custom skills for legal document generation
- Jira integration for feature tracking
- FastAPI backend with user authentication
- Automated deployment pipeline

```
# claude.md – Project context file
# Project: LegalDoc Generator
# Stack: FastAPI, PostgreSQL, Jinja2 templates
# Conventions: Black formatting, pytest, async views
# Key files: /app/main.py, /app/models/, /app/templates/
```

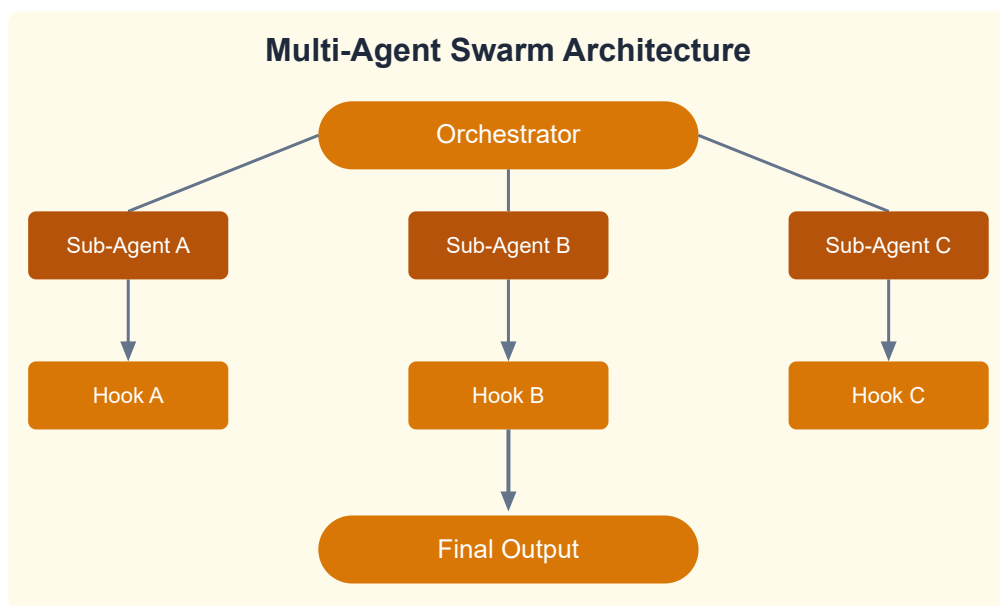
 **Exercise:** Set up Claude Code CLI in a project. Configure an MCP server for GitHub (issue management). Create a `claude.md` context file. Build a small SaaS feature end-to-end: create a Jira ticket from Claude, implement the feature, open a PR, and review the PR — all from the CLI.

# Vibe Engineering as an Expert

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## Multi-Agent Architectures

Expert-level vibe engineering uses swarms of sub-agents, each with specialized roles, orchestrated by hooks and custom commands.



## Sub-Agents & Hooks

```
// Claude swarm configuration
{
  "agents": {
    "frontend": {
      "role": "Frontend Developer",
      "context": ["/frontend/**", "tailwind.config.js"],
      "hooks": {
        "pre-commit": "lint:check",
        "post-deploy": "e2e:test"
      }
    },
    "backend": {
      "role": "Backend Developer",
      "context": ["/backend/**", "requirements.txt"],
      "hooks": {
        "pre-commit": "pytest",
        "post-merge": "migrate:db"
      }
    }
  }
}
```

## Controlled Chaos: Swarms

Swarms are groups of agents that work in parallel on different aspects of a problem. An orchestrator coordinates their outputs.

```
# Swarm orchestrator pattern
class SwarmOrchestrator:
    def __init__(self):
        self.agents = {
            "planner": Agent("plan"),
            "coder": Agent("code"),
            "tester": Agent("test"),
            "reviewer": Agent("review")
        }

    async def run(self, task):
        plan = await self.agents["planner"].run(task)
        code = await self.agents["coder"].run(plan)
        test_report = await self.agents["tester"].run(code)
        review = await self.agents["reviewer"].run(code, test_report)
        return review
```

## Cloud Sandboxes

Execute agents remotely in cloud sandboxes (Sprites.dev, Modal) for secure, scalable execution.

```
claude --sandbox sprites # Run in cloud sandbox
```

## Scale & SDKs

The Claude Agent SDK enables programmatic agent creation and orchestration for large codebases.

```
pip install claude-agent-sdk

from claude_agent import Agent, Swarm

agent = Agent(
    name="CodeReviewer",
    model="claude-sonnet-4-20250514",
    instructions="Review code for bugs, security, and style."
)

swarm = Swarm(agents=[agent], orchestrator="sequential")
result = swarm.run("Review the PR #42 changes")
```

## Advanced Swarms & Capstone

The Gastown orchestration pattern enables parallel agent execution with dynamic task assignment.

### Capstone: Trader Workstation

Build a complete trading platform using multi-agent swarms:

- **Data Agent** — Fetches market data from APIs
- **Analysis Agent** — Runs technical analysis
- **Risk Agent** — Assesses portfolio risk
- **Execution Agent** — Places trades via broker API
- **Reporting Agent** — Generates performance reports

 **Exercise:** Build a multi-agent swarm for a trading platform. Configure sub-agents for data, analysis, risk, and execution. Use hooks for pre-trade checks and post-trade logging. Deploy the orchestrator to a cloud sandbox. The final system should be able to: fetch real-time data, analyze trends, assess risk, and execute paper trades autonomously.

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## 4. Practice & Resources

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### Quick Reference Guide

| Task                  | Command / Tool                                     |
|-----------------------|----------------------------------------------------|
| Start Claude Code CLI | <code>claude</code>                                |
| Pipe prompt to Claude | <code>echo "prompt"   claude -p</code>             |
| Save checkpoint       | <code>claude --checkpoint save "name"</code>       |
| Load checkpoint       | <code>claude --checkpoint load "name"</code>       |
| YOLO mode             | <code>claude --yolo</code>                         |
| Plan before coding    | <code>/plan</code> command in Claude               |
| Review changes        | <code>/review</code> command in Claude             |
| MCP server config     | <code>claude.json</code> → <code>mcpServers</code> |
| Cloud sandbox         | <code>claude --sandbox sprites</code>              |
| Project context       | <code>claude.md</code> file in project root        |

### Useful Resources

- [Claude Code CLI Documentation](#)
- [MCP Specification](#)
- [Cursor IDE](#)
- [GitHub Copilot](#)
- [Sprites.dev — Cloud sandbox for AI coding](#)

### 3-Week Study Plan

| Week | Topic              | Activities                                                                  |
|------|--------------------|-----------------------------------------------------------------------------|
| 1    | Vibe Coding        | Setup Cursor/Claude Code, build Kanban app, learn context engineering       |
| 2    | Vibe Engineering   | Claude Code CLI mastery, MCP servers, Jira/GitHub integration, SaaS project |
| 3    | Expert Engineering | Multi-agent swarms, hooks, cloud sandboxes, trading platform capstone       |

### IDE / Tool Comparison

| Tool            | Cost        | Best For                            | AI Models      |
|-----------------|-------------|-------------------------------------|----------------|
| Cursor          | \$20/mo     | Daily coding with deep context      | Claude, GPT-4o |
| Copilot         | \$10/mo     | Enterprise teams, VS Code           | GPT-4o, Claude |
| Claude Code CLI | API usage   | Complex projects, agentic workflows | Claude         |
| Codex CLI       | API usage   | Terminal-native automation          | GPT-4o         |
| OpenCode        | Open source | Self-hosted, custom workflows       | Various        |

 **Final Capstone:** Build a complete multi-agent SaaS platform using all three levels: (1) vibe-code the initial prototype, (2) engineer it professionally with MCP servers, claude.md context, and CI/CD, (3) add expert-level multi-agent swarm orchestration for critical workflows. Deploy with cloud sandboxes and document the entire process.

# 5. References

## Glossary

| Term            | Definition                                                                     |
|-----------------|--------------------------------------------------------------------------------|
| Vibe Coding     | AI-native coding where you describe intent and AI generates code               |
| Claude Code CLI | Anthropic's terminal-based agentic coding tool                                 |
| MCP             | Model Context Protocol — standardized tool interface for AI                    |
| Context File    | <code>claude.md</code> or <code>agents.md</code> — project instructions for AI |
| Checkpoint      | Saved session state for rollback                                               |
| YOLO Mode       | Autonomous mode where AI runs commands without approval                        |
| Sub-Agent       | Specialized agent within a larger swarm                                        |
| Hook            | Automated action triggered by events (pre-commit, post-deploy)                 |
| Swarm           | Group of coordinated agents working in parallel                                |
| Sandbox         | Isolated cloud environment for safe AI code execution                          |

# Cheatsheet

```
# Claude Code CLI Cheatsheet

# Install
npm install -g @anthropic-ai/claude-code

# Start session
claude

# Quick prompt
echo "explain this" | claude -p

# Session controls
/plan      # Generate a plan
/review    # Review changes
/summarize # Get session summary
/checkpoint # Save state

# Flags
--yolo      # Autonomous mode
--model MODEL # Use specific Claude model
--resume    # Resume last session
--verbose   # Detailed output

# Config file (claude.json)
{
  "mcpServers": { "serverName": { "command": "node", "args": ["server.js"] } }
}

# Project context (claude.md)
# Place in project root with conventions, stack, and key files.
```

# Claude Model Comparison

| Model         | Best For                | Context |
|---------------|-------------------------|---------|
| Claude Haiku  | Fast, lightweight tasks | 200K    |
| Claude Sonnet | Balanced speed/quality  | 200K    |
| Claude Opus   | Complex reasoning, code | 200K    |

 **Congratulations!** You've completed the Claude Vibe Coding & Engineering course. You can now vibe-code prototypes, engineer production systems with Claude Code CLI, and orchestrate expert-level multi-agent swarms for complex applications.